

Resume Screening and Application Tracking System Report

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Introduction

What is the project?

This project develops an “Resume screening and ATS” , an application designed to assist job seekers in optimizing their resumes for specific job descriptions. It automates the process of extracting key information from a resume, identifying relevant skills, and calculating a compatibility score against a given job description.

Why is it useful?

In today's competitive job market, many companies use Applicant Tracking Systems (ATS) to filter resumes. This tool aims to help job applicants understand how well their resume aligns with a job's requirements, allowing them to tailor their applications more effectively. By providing a clear ATS match score and highlighting detected skills, it empowers users to improve their chances of passing initial screening stages.

Objectives

The primary objectives of the project are:

1. **Resume Text Extraction** : To accurately extract textual content from PDF resume files.
2. **Skill Identification**: To detect and list relevant technical and professional skills present in the resume.
3. **Job Description Processing**: To process and understand the key requirements and keywords from a job description.
4. **ATS Match Scoring**: To calculate a quantitative 'ATS Match Score' indicating the compatibility between a resume and a job description.
5. **User-Friendly Interface**: To provide an intuitive web-based interface (using Streamlit) for users to upload resumes, input job descriptions, and view analysis results.

Dataset Information

This project does not rely on a traditional pre-existing dataset in the sense of a fixed CSV or database. Instead, it processes user-provided inputs:

Source of the dataset: The 'dataset' for this application is dynamically provided by the user in two forms:

1. Resume File: A PDF document uploaded by the user, representing their resume.
2. Job Description: A plain text input provided by the user, outlining the requirements of a specific job.

Number of records and features: There are no fixed 'records' or 'features' in a tabular sense. Each interaction involves one resume and one job description.

Description of columns/inputs:

Resume (PDF): Contains free-form text, which is parsed to extract skills and overall content relevant to a job application.

Job Description (Text Area): Contains free-form text, which is parsed to extract keywords and context for comparison with the resume.

`skills\_db`: A predefined internal list of common technical skills used to identify skills within the resume text. This acts as a reference 'dataset' for skill extraction.

Methodology

The project employs the following steps and techniques:

1. PDF Text Extraction:

The `pdfplumber` library is used to open and extract text content from the uploaded PDF resume files. Each page's text is extracted and concatenated into a single string.

2. Skill Extraction:

- * A predefined list of `skills\_db` (e.g., Python, Java, SQL, Machine Learning) is maintained.
- * The extracted resume text is converted to lowercase.
- * The text is then iterated over, checking for the presence of each skill from the `skills\_db`. Identified skills are collected into a list.

3. ATS Match Score Calculation (Cosine Similarity):

- * Both the extracted resume text and the provided job description text are treated as separate documents.
- * The `CountVectorizer` from `scikit-learn` is utilized to transform these text documents into a matrix of token counts (term frequency-inverse document frequency, or TF-IDF, could also be used for more advanced weighting, but `CountVectorizer` provides a basic bag-of-words model).
- * Cosine Similarity** is then calculated between the vector representations of the resume and the job description. This metric measures the cosine of the angle between the two vectors, indicating their similarity. A higher cosine similarity implies greater textual alignment.
- * The similarity score is scaled to a percentage (0-100%) and rounded for presentation.

4. ****Streamlit User Interface:**

- * The entire process is encapsulated within a Streamlit application.
- * ``streamlit.file_uploader`` allows users to upload PDF resumes.
- * ``streamlit.text_area`` provides an input field for job descriptions.
- * ``streamlit.subheader`` and ``streamlit.write`` are used to display the extracted resume text, detected skills, and the final ATS match score to the user in an interactive web interface.